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Project Report

EXECUTIVE SUMMARY

The rapid growth of Artificial Intelligence and Computer Vision has introduced innovative solutions for the fashion industry. Technologies such as Virtual Try-On Systems, Fashion Recommendation Engines, Human Pose Estimation, and AR/VR experiences are transforming how customers interact with fashion products in digital environments.

This documentation presents a comprehensive study of AI-powered fashion technologies explored during the Raritone Summer Internship 2026. The research focuses on understanding the technologies, models, datasets, workflows, and methodologies used to create intelligent and personalized fashion platforms.

The documentation covers key domains including body detection, pose estimation, cloth segmentation, human parsing, 2D and 3D Virtual Try-On systems, recommendation engines, and immersive AR/VR fashion experiences. Various AI models and frameworks were analyzed to understand their applications, strengths, and contributions within the fashion technology ecosystem.

The study also examines the importance of fashion datasets and demonstrates how multiple AI components can be integrated into a complete workflow capable of delivering personalized recommendations and realistic garment visualizations.

The primary objective of this documentation is to provide a structured overview of modern fashion AI technologies and highlight their potential to improve online shopping experiences through personalization, automation, and intelligent decision-making.

The findings presented in this document demonstrate the growing role of Artificial Intelligence in shaping the future of fashion retail and digital commerce.

SYSTEM OVERVIEW

Introduction

The Fashion AI System is a conceptual framework that combines Artificial Intelligence, Machine Learning, Computer Vision, and Recommendation Technologies to enhance digital fashion experiences. The system is designed to assist users in visualizing garments, understanding product suitability, and receiving personalized fashion recommendations.

The overall workflow integrates multiple AI modules that work together to analyze user images, process garment information, and generate intelligent outputs.

Purpose of the System

The primary purpose of the system is to improve the online shopping experience by providing users with:

- Personalized fashion recommendations

- Virtual garment visualization
- Body analysis capabilities
- Interactive shopping experiences
- Intelligent decision support

The system aims to reduce uncertainty during online purchases and improve customer confidence.

Core Functional Areas

The Fashion AI System consists of several interconnected components.

Body Analysis Module

Responsible for:

- Human Detection
- Body Landmark Identification
- Body Structure Analysis

This module serves as the starting point of the workflow.

Pose Estimation Module

Responsible for:

- Body Keypoint Detection
- Skeleton Tracking
- Posture Analysis

The module helps the system understand user body positioning.

Segmentation Module

Responsible for:

- Cloth Segmentation
- Human Parsing
- Background Removal

The module separates important image regions for further processing.

Virtual Try-On Module

Responsible for:

- Garment Alignment
- Garment Warping

- Virtual Fitting

This module generates visual representations of clothing on users.

Recommendation Modul

Responsible for:

- Product Suggestions
- Outfit Recommendations
- Personalized Fashion Insights

The module enhances user engagement through intelligent recommendations.

System Inputs

The system may utilize the following inputs:

User Data

- User Images
- Fashion Preferences
- Clothing Selections

Fashion Data

- Garment Images
- Product Information
- Fashion Attributes

These inputs are processed by AI models to generate meaningful outputs.

System Outputs

The Fashion AI System generates:

- Virtual Try-On Results
- Personalized Recommendations
- Fashion Insights
- Product Suggestions
- Enhanced Shopping Experiences

These outputs assist users in making informed purchasing decisions.

High-Level Workflow

User Input

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Body Analysis

↓

Pose Detection

↓

Segmentation

↓

Virtual Try-On Processing

↓

Recommendation Generation

↓

Personalized Output

The workflow demonstrates how multiple AI modules collaborate to create intelligent fashion experiences.

Benefits of the System

The integrated system offers several advantages:

- Improved Product Visualization
- Enhanced Personalization
- Better Shopping Confidence
- Intelligent Recommendations
- Interactive User Experiences
- Increased Customer Engagement

These benefits contribute to a more effective and user-friendly fashion platform.

Summary

The Fashion AI System combines Computer Vision, Virtual Try-On Technologies, Recommendation Systems, and AI-driven analysis to create a comprehensive digital fashion solution. By integrating multiple technologies into a unified workflow, the system can provide personalized, interactive, and engaging experiences that support the future of intelligent fashion retail.

FASHION AI ECOSYSTEM

Overview

Modern fashion technology platforms are built using a combination of Artificial Intelligence, Machine Learning, Computer Vision, and Recommendation Systems. Rather than functioning independently, these technologies work together as an interconnected ecosystem to provide personalized and intelligent shopping experiences.

The Fashion AI Ecosystem represents the collaboration of multiple modules that collectively analyze user data, process fashion information, and generate meaningful outputs

Concept of the Fashion AI Ecosystem

A Fashion AI Ecosystem is a network of interconnected technologies that transform user inputs into personalized fashion experiences.

The ecosystem combines:

- Human Body Analysis
- Pose Estimation
- Image Segmentation
- Virtual Try-On Technologies
- Recommendation Systems
- AR/VR Experiences

Each component contributes specific functionality while supporting the overall workflow.

Major Components of the Ecosystem

1. User Interaction Layer

The User Interaction Layer acts as the entry point of the ecosystem.

Functions:

- User Registration
- Image Upload
- Garment Selection
- Preference Input

Purpose:

To collect information required for AI processing and personalization.

2. Computer Vision Layer

The Computer Vision Layer is responsible for understanding visual information.

Functions:

- Body Detection
- Landmark Detection
- Pose Estimation
- Image Analysis

Technologies Studied:

- OpenCV
- MediaPipe
- OpenPose

Purpose:

To analyze user images and extract meaningful body-related information.

3. Segmentation and Parsing Layer

This layer focuses on identifying clothing and body regions.

Functions:

- Human Parsing
- Cloth Segmentation
- Background Removal
- Image Masking

Technologies Studied:

- U2Net
- Detectron2

Purpose:

To improve garment fitting and virtual try-on quality.

4. Virtual Try-On Layer

The Virtual Try-On Layer generates digital garment visualizations.

Functions:

- Garment Alignment
- Garment Warping
- Clothing Visualization
- Virtual Fitting

Technologies Studied:

- VITON-HD
- HR-VITON

Purpose:

To provide realistic clothing previews before purchase.

5. Recommendation Layer

The Recommendation Layer generates personalized fashion suggestions.

Functions:

- Product Recommendations
- Outfit Suggestions
- Preference Analysis
- Trend-Based Recommendations

Approaches Studied:

- Content-Based Filtering
- Collaborative Filtering
- Hybrid Recommendation Systems

Purpose:

To enhance personalization and product discovery.

6. AR/VR Experience Layer

The AR/VR Layer focuses on immersive fashion experiences.

Functions:

- 3D Visualization
- Avatar Generation
- Virtual Product Exploration
- Interactive Shopping

Technologies Studied:

- Blender
- Unity
- Three.js

Purpose:

To provide advanced and engaging digital shopping environments.

Information Flow Across the Ecosystem

The Fashion AI Ecosystem operates through a continuous flow of information.

User Input

↓

Computer Vision Analysis

↓

Segmentation & Parsing

↓

Virtual Try-On Processing

↓

Recommendation Generation

↓

AR/VR Enhancement

↓

Final User Experience

This workflow ensures that all modules contribute to the final output.

Benefits of an Integrated Ecosystem

The integration of multiple AI technologies provides several advantages.

Benefits include:

- Personalized User Experiences
- Improved Product Visualization
- Better Shopping Decisions
- Increased Customer Engagement
- Enhanced Recommendation Accuracy
- Realistic Virtual Fitting

These benefits help create a more efficient and customer-centric fashion platform.

Importance in Modern Fashion Retail

The Fashion AI Ecosystem represents the future of digital fashion commerce.

Key Contributions:

- Supports personalization
- Improves customer confidence
- Enhances product discovery
- Reduces uncertainty in online shopping
- Creates interactive shopping experiences

As technology continues to evolve, integrated AI ecosystems will become increasingly important within the fashion industry.

Conclusion

The Fashion AI Ecosystem demonstrates how multiple Artificial Intelligence technologies can work together to create intelligent, personalized, and immersive fashion experiences. By combining Computer Vision, Virtual Try-On Systems, Recommendation Engines, and AR/VR technologies, fashion platforms can deliver advanced digital solutions that improve customer satisfaction and engagement.

The ecosystem provides a strong foundation for the future development of AI-powered fashion applications and next-generation online shopping platforms.

BODY DETECTION & ANALYSIS

Overview

Body Detection is one of the most fundamental components of modern fashion AI systems. Before performing advanced tasks such as pose estimation, virtual try-on generation, or recommendation analysis, the system must first identify and locate the user within an image.

During the internship, body detection technologies were researched to understand how Artificial Intelligence can recognize human body structures and extract meaningful information for further processing.

Purpose of Body Detection

The primary objective of body detection is to identify the presence and location of a person within an image or video.

The process helps the system:

- Locate the user
- Identify body regions
- Support pose estimation
- Prepare data for segmentation
- Improve virtual try-on accuracy

Without accurate body detection, subsequent AI modules may produce unreliable results.

Role in Fashion AI Systems

Body detection serves as the entry point for many fashion-related applications.

Applications include:

- Virtual Try-On Systems
- Body Measurement Analysis
- Pose Estimation
- Fashion Recommendation Platforms
- AR/VR Fashion Experiences

The accuracy of body detection directly influences the performance of downstream modules.

Body Detection Workflow

The body detection process follows a structured sequence of operations.

Input Image

↓

Image Preprocessing

↓

Human Detection

↓

Body Localization

↓

Region Identification

↓

Output for Further Analysis

This workflow ensures that the system accurately identifies the user before proceeding to advanced processing stages.

Image Preprocessing

Before body detection is performed, images often undergo preprocessing.

Activities include:

- Image resizing
- Brightness adjustment
- Noise reduction
- Quality enhancement

Benefits:

- Improved detection accuracy
- Better image consistency
- Enhanced model performance

Preprocessing helps prepare visual data for efficient analysis.

Human Detection

Human detection is the process of identifying whether a person is present within an image.

Functions:

- Person recognition
- Object classification
- Region identification

Purpose:

To distinguish human subjects from other objects present in the image.

Body Localization

Once a person is detected, the system identifies the location of the body within the image.

Functions:

- Region extraction
- Position identification
- Body boundary estimation

Benefits:

- Improved analysis accuracy
- Better landmark detection
- Efficient processing

Body localization provides the foundation for pose estimation and measurement analysis.

Body Region Analysis

Body region analysis involves identifying important areas of the human body.

Examples:

- Head Region
- Shoulder Region
- Torso Region
- Arm Regions
- Leg Regions

Purpose:

To understand body structure and support later stages of the workflow.

This information becomes valuable when aligning garments and estimating measurements.

Technologies Explored

Several Computer Vision technologies were researched to understand body detection systems.

OpenCV

Applications:

- Image Processing
- Object Detection
- Feature Extraction

MediaPipe

Applications:

- Body Landmark Detection
- Human Tracking
- Real-Time Analysis

These technologies provide efficient solutions for body analysis tasks.

Importance in Virtual Try-On Systems

Body detection plays a critical role in virtual fitting applications.

Benefits include:

- Accurate garment placement
- Better body understanding
- Improved fitting quality
- Enhanced personalization

The effectiveness of virtual try-on systems largely depends on the quality of body analysis.

Challenges Identified

Several challenges were observed during the research.

Challenges include:

- Variations in body posture
- Different lighting conditions
- Complex backgrounds
- Partial body visibility
- Image quality limitations

Addressing these challenges is important for improving system reliability.

Key Observations

The research resulted in several important observations:

- Accurate body detection improves overall workflow performance.
- Preprocessing contributes to better detection results.
- Body localization supports pose estimation.
- Human analysis improves personalization.
- Reliable body detection enhances virtual fitting accuracy.

These findings demonstrate the importance of body analysis within intelligent fashion systems.

Conclusion

Body Detection and Analysis form the foundation of AI-powered fashion applications. By identifying users, locating body regions, and preparing information for advanced processing, body detection enables technologies such as pose estimation, segmentation, virtual try-on systems, and recommendation engines.

The research highlighted the critical role of body analysis in creating accurate, personalized, and effective digital fashion experiences.

POSE ESTIMATION FRAMEWORK

Overview

Pose Estimation is a Computer Vision technique that enables Artificial Intelligence systems to identify and track human body landmarks within images and videos. It plays a crucial role in fashion technology applications because it helps systems understand body posture, movement, and structural alignment.

During the internship, pose estimation technologies were studied to understand how body keypoints can be utilized for garment fitting, body analysis, and virtual try-on systems.

Purpose of Pose Estimation

The primary objective of pose estimation is to detect and analyze the position of different body parts.

The framework helps the system:

- Identify body landmarks
- Understand body posture
- Track body movements
- Support garment alignment
- Improve fitting accuracy

Pose estimation acts as a bridge between body detection and virtual try-on generation.

Importance in Fashion Technology

Fashion AI systems rely heavily on pose information to create realistic and personalized experiences.

Applications include:

- Virtual Try-On Systems
- Body Measurement Analysis
- Garment Alignment
- Motion Tracking
- AR/VR Fashion Experiences

Accurate pose information helps ensure that garments are positioned correctly on the user's body.

Human Body Landmarks

Pose estimation frameworks identify important body landmarks known as keypoints.

Common landmarks include:

- Head
- Neck
- Shoulders
- Elbows
- Wrists
- Hips
- Knees
- Ankles

These landmarks collectively form a skeletal representation of the human body.

Pose Estimation Workflow

The process follows a structured workflow.

Input Image

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Body Detection

↓

Landmark Detection

↓

Skeleton Construction

↓

Pose Analysis

↓

Body Structure Information

This workflow enables the system to understand body posture and positioning.

Landmark Detection

Landmark detection is the process of identifying specific body points.

Functions:

- Keypoint Identification
- Position Tracking
- Body Mapping

Benefits:

- Accurate body analysis
- Improved garment placement
- Better measurement estimation

Landmarks provide valuable information for understanding body structure.

Skeleton Construction

Once landmarks are identified, they are connected to create a skeletal representation.

Purpose:

- Visualize body posture
- Understand body orientation
- Analyze movement patterns

Benefits:

- Enhanced body interpretation
- Improved fitting support
- Better motion analysis

Skeleton tracking is widely used in fashion, healthcare, and sports applications.

Technologies Studied

Several pose estimation frameworks were researched during the internship.

MediaPipe Pose

Applications:

- Real-Time Pose Detection
- Landmark Tracking
- Skeleton Visualization

Advantages:

- Fast processing speed
- Lightweight architecture
- Easy integration

MediaPipe is suitable for applications requiring real-time body analysis.

OpenPose

Applications:

- Human Pose Estimation
- Multi-Person Tracking
- Detailed Landmark Detection

Advantages:

- High accuracy
- Detailed body keypoints
- Robust pose analysis

OpenPose is commonly used in advanced pose estimation tasks.

Role in Virtual Try-On Systems

Pose estimation is one of the most important stages of the virtual try-on workflow.

Functions:

- Determines body posture
- Supports garment alignment
- Improves fitting realism
- Enables body-aware garment placement

Without accurate pose estimation, virtual fitting systems may generate unrealistic outputs.

Body Measurement Support

Pose estimation frameworks can assist in estimating body proportions and measurements.

Possible Measurements:

- Shoulder Width
- Body Height
- Torso Proportions
- Limb Length Estimation

Benefits:

- Personalized fitting
- Better size recommendations
- Enhanced user experience

This capability contributes to intelligent fashion applications.

Challenges in Pose Estimation

Several challenges were identified during the research.

Challenges include:

- Occluded body parts
- Complex body poses
- Image quality variations
- Lighting conditions
- Multi-person environments

Addressing these challenges is important for improving landmark detection accuracy.

Key Findings

The study of pose estimation resulted in several important observations.

Findings:

- Accurate landmark detection improves garment fitting.
- Skeleton tracking enhances body understanding.
- Pose information supports measurement estimation.
- Real-time tracking enables interactive applications.
- Pose estimation significantly improves virtual try-on quality.

These findings highlight the importance of pose analysis in intelligent fashion systems.

Conclusion

The Pose Estimation Framework serves as a critical component of modern fashion AI applications. By detecting body landmarks, constructing skeletal structures, and analyzing body posture, pose estimation provides the information required for garment alignment, measurement analysis, and virtual fitting.

The research demonstrated that accurate pose estimation significantly improves the performance of Virtual Try-On systems and contributes to creating realistic, personalized, and engaging fashion experiences.

SEGMENTATION & HUMAN PARSING FRAMEWORK

Overview

Segmentation and Human Parsing are essential Computer Vision techniques used in modern fashion AI systems. These technologies enable the system to identify and separate different regions within an image, such as clothing, body parts, and background elements.

During the internship, segmentation workflows and human parsing models were researched to understand their role in improving Virtual Try-On systems, garment visualization, and image analysis.

The study highlighted how segmentation contributes to creating realistic and accurate fashion experiences.

Understanding Segmentation

Image Segmentation is the process of dividing an image into meaningful regions.

The objective is to identify and isolate specific objects or areas within an image.

Examples:

- Clothing Regions
- Human Body Regions
- Background Areas
- Fashion Accessories

Segmentation helps AI systems focus on relevant image components while ignoring unnecessary information.

Understanding Human Parsing

Human Parsing is a specialized form of segmentation that identifies different body parts and clothing categories separately.

Examples of Parsed Regions:

- Hair
- Face
- Upper Clothing
- Lower Clothing

- Arms
- Legs
- Shoes

This detailed understanding improves the accuracy of virtual fitting systems.

Importance in Fashion AI

Segmentation and human parsing play a crucial role in fashion-related applications.

Applications include:

- Virtual Try-On Systems
- Cloth Segmentation
- Garment Extraction
- Background Removal
- Fashion Image Analysis

These technologies help improve realism and fitting accuracy.

Segmentation Workflow

The segmentation process follows a structured workflow.

Input Image

↓

Image Preprocessing

↓

Region Identification

↓

Segmentation Mask Generation

↓

Human Parsing

↓

Separated Image Components

This workflow allows the system to identify and isolate relevant image regions.

Cloth Segmentation

Cloth segmentation focuses specifically on identifying clothing areas within an image.

Functions:

- Garment Identification
- Clothing Extraction
- Fashion Item Separation

Benefits:

- Improved garment analysis
- Better virtual fitting
- Enhanced clothing visualization

Cloth segmentation is one of the most important steps in Virtual Try-On workflows.

Background Removal

Background removal separates the user and clothing from unnecessary image elements.

Functions:

- Object Isolation
- Cleaner Image Generation
- Focused Processing

Benefits:

- Better segmentation quality
- Improved visual appearance
- Enhanced fitting accuracy

Removing background distractions improves overall system performance.

Segmentation Models Studied

Several segmentation models were researched during the internship.

U2Net

Applications:

- Background Removal
- Image Masking
- Cloth Segmentation

Advantages:

- High segmentation quality
- Efficient object extraction

- Accurate masking

U2Net is commonly used for image preprocessing and garment extraction tasks.

Detectron2

Applications:

- Human Parsing
- Object Detection
- Region Segmentation

Advantages:

- Detailed segmentation outputs
- Advanced recognition capabilities
- Strong parsing performance

Detectron2 provides detailed information about body and clothing regions.

Role in Virtual Try-On Systems

Segmentation serves as a critical component of virtual fitting workflows.

Functions:

- Extracts clothing regions
- Separates body components
- Supports garment alignment
- Improves fitting quality

Without accurate segmentation, garments may not align properly with the user's body.

Challenges in Segmentation

Several challenges were identified during the research.

Challenges include:

- Complex backgrounds
- Similar color patterns
- Clothing overlaps
- Occluded body regions
- Variations in lighting

These challenges can affect segmentation accuracy and overall system performance.

Key Findings

The study resulted in several important observations.

Findings:

- Accurate segmentation improves virtual try-on realism.
- Human parsing enhances body understanding.
- Background removal improves image quality.
- Segmentation quality directly impacts fitting accuracy.
- Detailed region identification supports personalization.

These findings demonstrate the importance of segmentation in fashion AI systems.

Benefits of Human Parsing and Segmentation

The integration of segmentation and parsing technologies provides several advantages.

Benefits:

- Better garment extraction
- Improved fitting quality
- Enhanced image processing
- Realistic virtual visualization
- Increased personalization

These benefits contribute significantly to user satisfaction and shopping confidence.

Conclusion

The Segmentation and Human Parsing Framework is a vital component of modern fashion technology systems. By identifying clothing regions, separating body parts, and removing unnecessary background information, these technologies enable more accurate and realistic Virtual Try-On experiences.

The research demonstrated that segmentation quality directly influences the effectiveness of garment fitting, visualization, and personalization, making it one of the most important stages within the Fashion AI workflow.

2D VIRTUAL TRY-ON ARCHITECTURE

Overview

2D Virtual Try-On Architecture is a Computer Vision-based framework that enables users to visualize garments on their images without physically wearing them. The architecture combines body analysis, pose estimation, segmentation, garment alignment, and image generation techniques to create realistic clothing visualizations.

During the internship, research was conducted on the architecture, workflow, and components of modern 2D Virtual Try-On systems to understand how Artificial Intelligence can improve online fashion shopping experiences.

Purpose of 2D Virtual Try-On Systems

The primary objective of a 2D Virtual Try-On system is to allow users to preview garments before purchasing them.

The system aims to:

- Improve shopping confidence
- Reduce uncertainty during purchases
- Enhance garment visualization
- Increase user engagement
- Support personalized fashion experiences

These benefits make Virtual Try-On technology a valuable feature for modern fashion platforms.

Architecture Components

The 2D Virtual Try-On Architecture consists of several interconnected modules.

User Image Input

The process begins with the user uploading an image.

Input Requirements:

- Clear body visibility
- Good image quality
- Proper lighting conditions

Purpose:

To provide visual data for AI analysis.

Body Detection Module

The Body Detection Module identifies the user within the uploaded image.

Functions:

- Human Identification
- Body Localization
- Region Detection

Purpose:

To establish the foundation for further image analysis.

Pose Estimation Module

Pose estimation analyzes body posture and identifies body landmarks.

Functions:

- Landmark Detection
- Skeleton Tracking
- Body Structure Analysis

Technologies Studied:

- MediaPipe
- OpenPose

Purpose:

To provide positioning information for garment fitting.

Segmentation Module

Segmentation separates body and clothing regions from the image.

Functions:

- Cloth Segmentation
- Human Parsing
- Background Removal

Technologies Studied:

- U2Net
- Detectron2

Purpose:

To improve fitting accuracy and garment placement.

Garment Processing Module

This module prepares the selected garment for virtual fitting.

Functions:

- Garment Extraction
- Image Normalization
- Garment Analysis

Purpose:

To ensure that clothing images are suitable for alignment and fitting.

Garment Alignment Module

Garment alignment positions the selected garment according to body posture and structure.

Functions:

- Position Adjustment
- Landmark Mapping
- Shape Matching

Purpose:

To ensure realistic clothing placement.

Garment Warping Module

Garment warping adjusts the garment shape to match different body structures.

Functions:

- Shape Adaptation
- Geometry Transformation
- Fitting Enhancement

Purpose:

To create realistic fitting results.

Virtual Try-On Generation Module

This module generates the final virtual fitting output.

Functions:

- Clothing Overlay
- Image Synthesis
- Output Generation

Technologies Studied:

- VITON-HD
- HR-VITON

Purpose:

To create realistic garment visualizations.

Architectural Workflow

The overall workflow can be represented as follows:

User Image

↓

Body Detection

↓

Pose Estimation

↓

Segmentation

↓

Garment Selection

↓

Garment Alignment

↓

Garment Warping

↓

Virtual Try-On Generation

↓

Final Output

This architecture demonstrates how multiple AI modules collaborate to generate a virtual fitting experience.

Technologies Supporting the Architecture

Several technologies contribute to the effectiveness of the architecture.

Computer Vision

Supports:

- Image Analysis
- Body Detection
- Segmentation

Deep Learning

Supports:

- Landmark Detection

- Garment Fitting
- Image Generation

Machine Learning

Supports:

- Pattern Recognition
- Personalization
- Recommendation Integration

Together, these technologies create an intelligent virtual fitting framework.

Advantages of the Architecture

The 2D Virtual Try-On Architecture offers several benefits.

Advantages:

- Improved product visualization
- Better shopping confidence
- Enhanced personalization
- Increased customer engagement
- Reduced uncertainty before purchase

These benefits contribute to better online shopping experiences.

Challenges Identified

Several challenges were observed while studying Virtual Try-On architectures.

Challenges:

- Pose variation handling
- Segmentation accuracy
- Garment alignment complexity
- Realistic fitting generation
- Dataset limitations

Addressing these challenges remains an active area of research.

Key Observations

The research produced several important findings.

Findings:

- Pose estimation improves garment placement.
- Segmentation enhances fitting realism.
- Garment warping improves visualization quality.
- Image preprocessing supports better outputs.
- Multiple AI modules must work together for effective results.

These observations demonstrate the complexity and effectiveness of modern Virtual Try-On systems.

Conclusion

The 2D Virtual Try-On Architecture represents a powerful application of Artificial Intelligence and Computer Vision within the fashion industry. By integrating body detection, pose estimation, segmentation, garment alignment, and image generation techniques, the architecture enables realistic virtual fitting experiences that enhance customer confidence and engagement.

The research highlighted the growing importance of Virtual Try-On technologies in modern e-commerce and their potential to redefine digital fashion experiences.

3D VIRTUAL TRY-ON ARCHITECTURE

Overview

3D Virtual Try-On Architecture represents the next generation of digital fashion technology. Unlike traditional 2D systems that rely on image overlays, 3D Virtual Try-On systems create interactive digital representations of users and garments within a three-dimensional environment.

The architecture combines Artificial Intelligence, Computer Vision, 3D Modeling, Body Reconstruction, and Cloth Simulation technologies to provide highly realistic and immersive fashion experiences.

During the internship, research was conducted to understand the architecture, components, workflow, and technologies that support modern 3D Virtual Try-On systems.

Purpose of 3D Virtual Try-On Systems

The primary goal of 3D Virtual Try-On systems is to provide users with a realistic visualization of how garments may appear on their bodies.

Objectives include:

- Personalized garment visualization
- Improved fitting representation
- Enhanced customer confidence
- Interactive shopping experiences
- Immersive fashion exploration

These systems aim to bridge the gap between physical and digital shopping experiences.

Architecture Components

The architecture consists of multiple interconnected modules that work together to create realistic virtual fitting experiences.

User Data Input Module

The process begins with collecting user information.

Inputs:

- User Images
- Body Measurements
- Fashion Preferences

Purpose:

To provide information required for avatar creation and fitting analysis.

Body Reconstruction Module

This module analyzes the user and generates a digital body representation.

Functions:

- Body Shape Analysis
- Body Structure Identification
- Measurement Estimation

Purpose:

To create a personalized body model for virtual fitting.

Avatar Generation Module

The Avatar Generation Module creates a 3D digital representation of the user.

Functions:

- Avatar Creation
- Appearance Representation
- Personalization

Purpose:

To serve as the foundation for garment visualization.

Benefits:

- Improved personalization
- Better user representation
- Enhanced realism

Body Mesh Generation Module

This module converts body information into a three-dimensional mesh.

Functions:

- Mesh Construction
- Surface Modeling
- Body Geometry Creation

Purpose:

To provide a realistic structure for garment fitting.

Benefits:

- Accurate body representation
- Improved fitting quality
- Enhanced visualization

Garment Modeling Module

The Garment Modeling Module prepares clothing assets for simulation.

Functions:

- Garment Creation
- Fabric Representation
- Clothing Structure Modeling

Purpose:

To create realistic digital garments.

Cloth Simulation Module

This module simulates the physical behavior of garments.

Functions:

- Fabric Movement
- Wrinkle Generation
- Stretch Simulation
- Cloth Dynamics

Purpose:

To improve realism and fitting accuracy.

Benefits:

- Natural garment behavior
- Better visualization
- Enhanced customer understanding

Motion Tracking Module

Motion tracking enables interaction between users and avatars.

Functions:

- Movement Detection
- Pose Tracking
- Animation Control

Purpose:

To create dynamic virtual experiences.

Benefits:

- Interactive fitting
- Real-time feedback
- Improved immersion

Rendering Module

The rendering system generates the final visual output.

Functions:

- Scene Visualization
- Lighting Processing
- Real-Time Rendering

Purpose:

To create realistic virtual fitting experiences.

3D Virtual Try-On Workflow

The workflow follows a structured architecture.

User Input

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Body Analysis

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Body Reconstruction

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Avatar Generation

↓

Body Mesh Creation

↓

Garment Modeling

↓

Cloth Simulation

↓

Motion Tracking

↓

Rendering

↓

3D Virtual Try-On Output

This workflow demonstrates how multiple technologies collaborate to create immersive fashion experiences.

Technologies Explored

Several technologies commonly used in 3D fashion systems were researched.

Blender

Applications:

- 3D Modeling
- Avatar Design
- Cloth Simulation

Unity

Applications:

- Real-Time Rendering
- AR/VR Experiences
- Interactive Applications

Three.js

Applications:

- Web-Based 3D Visualization
- Interactive Fashion Platforms

SMPL Models

Applications:

- Human Body Modeling
- Avatar Generation
- Body Reconstruction

These technologies collectively support modern virtual fashion ecosystems.

Integration with AR and VR

The architecture can be further enhanced through AR and VR technologies.

AR Integration

Applications:

- Mobile-Based Try-On
- Product Visualization
- Interactive Fashion Experiences

VR Integration

Applications:

- Virtual Fashion Stores
- Immersive Shopping
- Interactive Product Exploration

Benefits:

- Enhanced engagement
- Better product understanding
- Immersive user experiences

Advantages of 3D Architecture

The research identified several advantages of 3D Virtual Try-On systems.

Benefits:

- Improved garment visualization
- Better fitting representation
- Personalized experiences

- Interactive user engagement
- Realistic cloth behavior
- Immersive shopping environments

These advantages make 3D systems highly valuable for future fashion platforms.

Challenges Identified

Several challenges were observed during the research.

Challenges:

- Complex avatar generation
- Computational requirements
- Cloth simulation accuracy
- Real-time rendering performance
- Hardware limitations

Addressing these challenges is important for widespread adoption.

Key Findings

The study resulted in several important observations.

Findings:

- 3D avatars improve personalization.
- Body reconstruction enhances fitting accuracy.
- Cloth simulation improves realism.
- Motion tracking enables interaction.
- AR/VR technologies enhance user engagement.
- 3D systems provide richer experiences than traditional methods.

These findings demonstrate the growing importance of immersive technologies in fashion applications.

Conclusion

The 3D Virtual Try-On Architecture represents a significant advancement in digital fashion technology. By integrating body reconstruction, avatar generation, cloth simulation, motion tracking, and rendering systems, the architecture creates realistic and immersive shopping experiences.

The research demonstrated that 3D technologies have the potential to transform online fashion retail by improving personalization, visualization, and customer engagement while laying the foundation for future AR/VR-based fashion ecosystems.